How to Study Boundary Phenomena: English Reciprocals and the Limits of Categorization

Abstract

English reciprocals (*each other* and *one another*) are a small-n boundary case: there are only two items, but both show persistent evidence of affiliation with pronouns and with compound determinatives. Instead of forcing a binary classification, this paper measures the STABILITY OF DIAGNOSTIC AMBIGUITY. Using a 155-property profile spanning morphology, syntax, semantics, and phonology, I locate the reciprocals in grammatical space and evaluate them with five lenses: distance measurement with a quasiswap reference distribution, specification-curve analysis, comparison-group checks, mixture calibration, and calibration against known categories. Across analyses, reciprocals occupy a stable boundary position between canonical pronouns and compound determinatives. I argue that this pattern is compatible with a homeostatic property-cluster interpretation of grammatical categories, and I offer the method as a transparent template for other small-n categorization problems.

Keywords: boundary phenomena; English reciprocals; grammatical categories; pronouns; determinatives

1 Introduction

When linguists categorize boundary phenomena – items that straddle categorical divisions – we face a compound problem. English reciprocals pose this challenge: not only are there just two (*each other* and *one another*), but they also exhibit properties of both pronouns and compound determinatives.

How can we rigorously investigate categorical ambiguity when our entire dataset consists of two boundary-dwelling items? Small samples of clear cases pose little difficulty – English has few words that satisfy the classic auxiliary diagnostics (Sag et al. 2020), but their status as verbs is uncontroversial. The challenge emerges when scarce data meets genuine categorial ambiguity, precisely the items linguists argue about.[1]

The traditional solution is to pick diagnostic tests and argue about them. Can it be a subject? Does it inflect for case? But this invites what Croft (2001) calls "methodological opportunism" – consciously or not, we select tests that support our preferred analysis. When we have only two items to classify, this cherry-picking becomes especially tempting.

This paper proposes a different approach: instead of seeking a decisive verdict, I measure the STABILITY OF DIAGNOSTIC AMBIGUITY. The interesting question isn't "which category?" but "how stable is the apparent boundary position under different measurement choices?"

If grammatical categories are homeostatic property clusters (Miller 2021) whose observable behaviour is boundary-like rather than all-or-nothing, then some items should consistently appear ambiguous no matter how we analyze them.[2]

On this homeostatic property-cluster (HPC) view, categories are stabilized by partially distinct mechanisms that sustain overlapping bundles of properties.[3]

This yields concrete, testable predictions – not arbitrary methodological desiderata but consequences of the theoretical framework itself: (E1) a boundary item should show invari-

[1] Such as the categorial status of *to* (Levine 2012).

[2] For word-kinds, Miller (2021) distinguishes internal cognitive and external social mechanisms. In applying this template to grammatical categories, I treat the feature matrix as an operationalization of the cluster and take the relevant mechanisms to include morphological realization rules, agreement/case systems, entrenched distributional patterns, grammaticalization pathways, and community norms.

[3] Abbreviations used below are as follows: CGEL = *The Cambridge Grammar of the English Language*; HPC = homeostatic property cluster; MCA = Multiple Correspondence Analysis; IDF = inverse document frequency; k-NN = k-nearest neighbours; PCoA = Principal Coordinates Analysis; PPC = posterior predictive check; PSIS-LOO = Pareto-smoothed importance-sampling leave-one-out; and Type S / Type M = sign / magnitude errors.

ance of its boundary position across reasonable analytic choices (distance metric, regularization, feature ablations); (E2) different feature families may exert opposed pulls if distinct mechanisms sustain partly conflicting bundles (cross-dimensional tension); and (E3) clear anchors should calibrate cleanly under the same lenses.

(E4) a one-parameter blend fit to anchor profiles should place boundary items near parity; and (E5) boundary diagnoses should persist when compared to reference distributions that preserve the instrument's marginals. The analyses below are organized to test these predictions directly.

I demonstrate this approach using English reciprocals as a test case. Prior descriptive work has noted their uncertain status, sitting between canonical pronouns and compound determinatives (Huddleston & Pullum 2002: henceforth CGEL, Ch. 5 §9.6): items like *somebody* and *anybody* that realize fused determiner–head functions (Payne, Huddleston & Pullum 2007).

The fusion-of-functions architecture predicts that morphologically complex items can belong to the determinative category while serving hybrid syntactic roles, making reciprocals a useful probe for boundary behaviour.

The approach has five components, each addressing a specific challenge in small-n categorization. First, I use a 155-feature profile: 155 binary properties covering morphology, syntax, semantics, and phonology and apply this to 138 words that form the universe of pronouns and determinatives as defined by CGEL.

This makes theoretical commitments explicit rather than hiding them across cherry-picked diagnostics. The logic is analogous to mapping vowel space: instead of relying on one or two diagnostics, I measure a broad set of properties and inspect their joint pattern.

Second, I visualize the high-dimensional feature space to build intuition about reciprocals' position. Figure 1 uses Multiple Correspondence Analysis (MCA) (Greenacre 2017), an ordination technique that arranges 155-dimensional data in 2 dimensions while preserving distances between items, much like reducing vowel measurements to an F1–F2 plot.

MCA naturally handles sparse binary data through χ^2 geometry.[4] But this is purely illustrative. For actual measurement, I use Jaccard distance for reasons explained below. All inferential contrasts operate on these full-dimensional Jaccard distances (with other variants as checks), never on the low-dimensional projections.

Third, I compare the observed contrast to a constrained reference distribution. With only two reciprocals, any one summary could be misleading. So I scramble the data 5,000 times using the quasiswap algorithm (Miklós & Podani 2004) while preserving its basic structure. I ask whether row- and column-preserving scramblings typically produce a determinative-ward distance contrast as large as the one observed here. This doesn't by itself establish boundary status; it checks whether one component of the overall pattern is unusual under a benchmark that preserves the instrument's marginals.

By this point, I've already made several analytical choices: Jaccard distance, specific reference items, and a particular test statistic. Each choice is a fork in the path.

So fourth, I map what Gelman & Loken (2013) call the "garden of forking paths" by varying every reasonable alternative and showing all results. Different distance metrics, different comparison groups, and different feature sets – I try them all and display the full spectrum. If the conclusion depends heavily on arbitrary choices, it should be immediately apparent.

Fifth, I calibrate against known structure from both directions. Bottom-up: can my methods correctly recover the categories from the CGEL framework? Top-down: what category mixture would generate the reciprocal pattern we observe? If the machinery can't identify clear pronouns as pronouns, the whole enterprise fails.

Two limitations constrain this entire analysis. First, everything depends on the hand-coded feature matrix – if the instrument is flawed, so are the conclusions. Second, with $n = 2$, statistical power is inherently limited. I address these constraints through transparency and

[4] MCA's χ^2 distance weights deviations from independence: rare shared features contribute more to distance calculations than common absences, making it appropriate for grammatical matrices where most features are absent for most words.

multiple robustness checks, but I can't eliminate them.

The obvious objection is that using elaborate statistics with only two reciprocals is futile. A phonetic calibration analogy dissolves that worry. Once a language's vowel space is mapped, a single token can be located relative to established categories without collecting more tokens of that type.

A lone production near the /i/–/ɪ/ boundary can be characterized as boundary-dwelling – but the token *is* either /i/ or /ɪ/; there's a fact of the matter. Our acoustic measurements simply can't resolve which. A lone instance of the front rounded vowel [y] in a loan name can be placed outside English's native inventory by its F1–F2–F3 coordinates. The 155-feature matrix plays the same role here: it's the calibrated space, with the same resolution limits.

The analyses below don't try to show that reciprocals constitute their own category (which $n = 2$ cannot support). They locate them in the calibrated space defined by clear PRONOUN and COMPOUND DETERMINATIVE anchors, compare one distance contrast to a row- and column-preserving reference distribution, and use mixture calibration and specification curves to check whether the broader interface reading is stable across analytic choices.

The payoff isn't a definitive recategorization – it's a template for investigating boundary phenomena honestly. When the inventory itself is fixed, we can still be transparent about what the available evidence shows. The methods are deliberately portable: other researchers can apply this same approach to their own small-n categorization problems with minimal adaptation.

2 The Challenge of Measuring Grammatical Similarity

When we ask whether reciprocals are pronouns or determinatives, we're really asking how to measure grammatical similarity. Traditional approaches pick a handful of diagnostic tests and declare success when items pattern together. That invites methodological opportunism. To avoid this trap, I use the calibrated instrument from the introduction: a 155-feature matrix that treats each word form as a point in a high-dimensional space whose axes are grammatical properties.

The features come from earlier coding work on this dataset. I keep only the 155 that apply to more than one word form – this filters out idiosyncratic properties unique to individual items, leaving features that actually help compare categories.

They span morphology (66), syntax (50), semantics (36), and phonology (3). With binary coding, every word form becomes a 155-bit vector.

Again, the vowel analogy is useful. Once you have mapped a language's vowel space, you can place any new token relative to the established categories. Here, the 155 features are the acoustic dimensions; canonical pronouns and compound determinatives are the reference categories; the 2D ordination is like an F1–F2 plot for visual orientation; and Jaccard distance stands in for perceptual distance.

I use Jaccard because in sparse binary data, shared absences tell you almost nothing – the fact that neither *each other* nor *the* can be plural doesn't reveal much about their relationship. What matters are shared presences and conflicts. Jaccard ignores the uninformative zeros and focuses on what counts. The MCA plot in Figure 1 is just for intuition; all the actual contrasts use full 155-dimensional distances.

For comparison, I use two predeclared sets. First, canonical pronouns like *he*, *her*, and *herself*. Second, the compound determinatives from Huddleston & Pullum (2002: Ch. 5 §9.6) – items like *anybody* and *everyone* that realize fused determiner-head functions (Payne, Huddleston & Pullum 2007). If morphologically complex reciprocals pattern with anything, it's probably with these hybrids rather than with simple pronouns.

Table 1 gives a compact view of the kinds of evidence that feed the matrix. The quantitative sections don't replace these diagnostics; they aggregate them, check their joint stability, and show how much the overall boundary reading depends on metric choice, feature family, and comparator choice.

From here the pipeline is straightforward: plot for orientation, compute distances, build

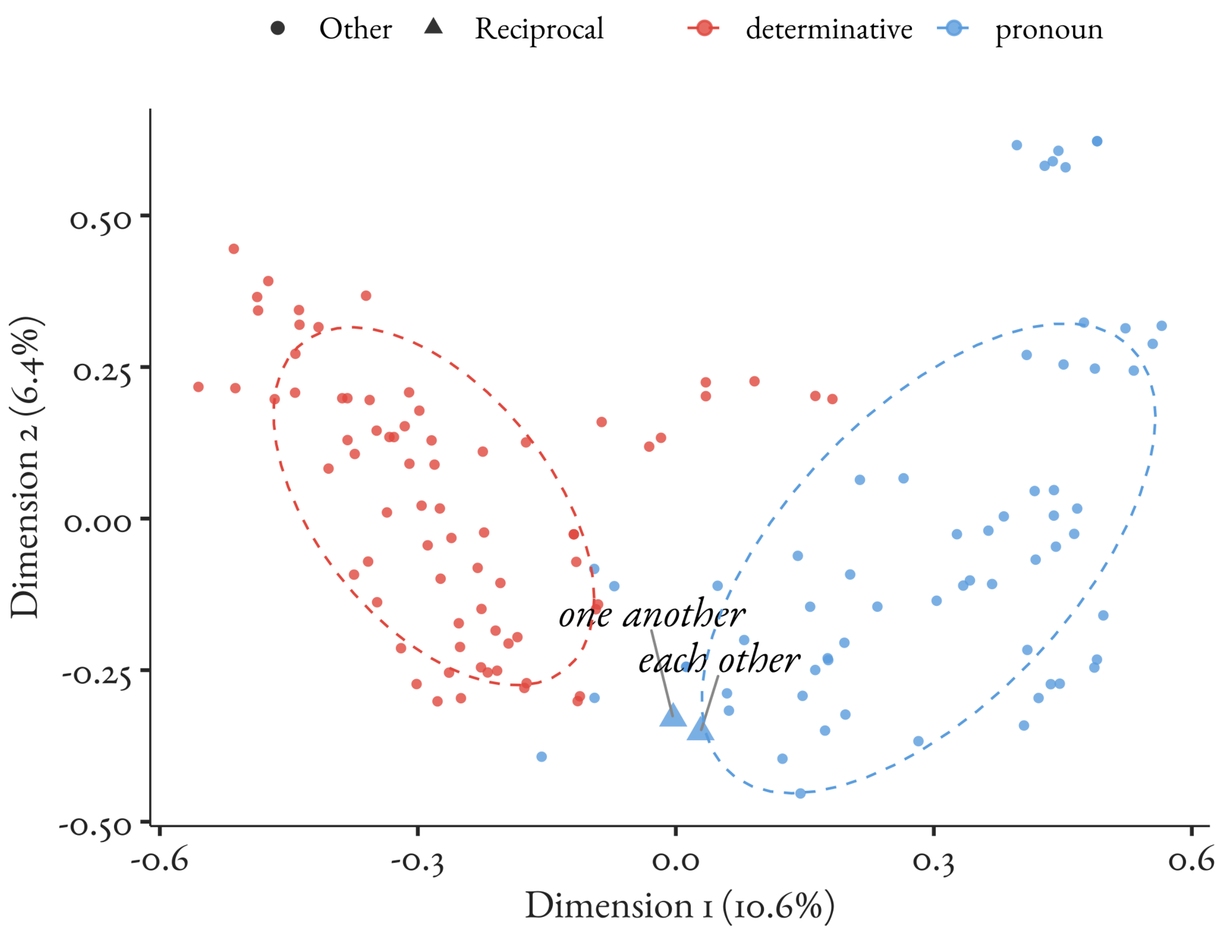


Figure 1: Reciprocals in grammatical feature space visualized with Multiple Correspondence Analysis. Pronouns (cyan) and determinatives (red) form regions; the compound determinatives sit at the interface; *each other* and *one another* (triangles) fall in that same interface.

a constrained reference distribution for one contrast, vary specifications to check robustness, and calculate mixture weights to summarize position. Each step tackles a specific challenge while keeping the choices transparent.

Diagnostic	Illustration	Canonical pronouns	Compound determinatives	Reciprocals	Main pull
Distinct case paradigm	*he/him/his*	yes	no	no	determinative-ward
Compound morphology	*somebody*, *each other*	no	yes	yes	determinative-ward
Pronoun-like anaphoric NP substitution	standalone anaphoric NP use	yes	no	yes	pronoun-ward
Fused determiner–head use	*somebody arrived*	no	yes	no	pronoun-ward
Reciprocal-style co-argument dependency	*They liked each other*	no	no	yes	distinctive
Subject distribution	*He arrived*, *Somebody arrived*	yes	yes	no	distinctive
Reciprocal semantics	mutual interpretation	no	no	yes	distinctive
Distinct pronominal genitive paradigm vs. clitic genitive	*his*, *somebody's*, *each other's*	yes	no	mixed	mixed

Table 1: Representative diagnostics encoded in the 155-property matrix. The full matrix contains many more items and finer distinctions; this subset shows the kinds of evidence that pull reciprocals toward pronouns, toward compound determinatives, or away from both.

3 Benchmarking the Distance Contrast Against a Constrained Reference Distribution

Having established how to measure distances, I next turned to a narrower question: is the observed determinative-ward contrast unusual under a constrained reference model? I address that question with row- and column-preserving quasiswap scramblings.

Each word in the dataset has a certain number of properties (e.g., *each other* has 23), and each property applies to a certain number of words (e.g., `inflects for case` applies to 75). These totals reflect real facts about English – that pronouns tend to have case distinctions, that most words can't be gradable, and so on.

But what if we kept these basic facts while scrambling which specific words have which specific properties while holding the totals constant? If the observed determinative-ward contrast is unusual under such scrambling, that tells us the contrast isn't a routine byproduct of the matrix's row and column totals. If it were common, we would treat it as a weak feature of the instrument rather than as informative structure.

The quasiswap algorithm (Miklós & Podani 2004) performs exactly this controlled scrambling. It rearranges the feature matrix through thousands of small swaps, always preserving how many features each word has and how many words have each feature. The result is a reference distribution for the observed Δ contrast under a benchmark that keeps the matrix marginals fixed. This doesn't tell us whether reciprocals are "really" intermediate. It asks a narrower question: whether the determinative-ward component of the reciprocal profile is unusual under that constrained scrambling.

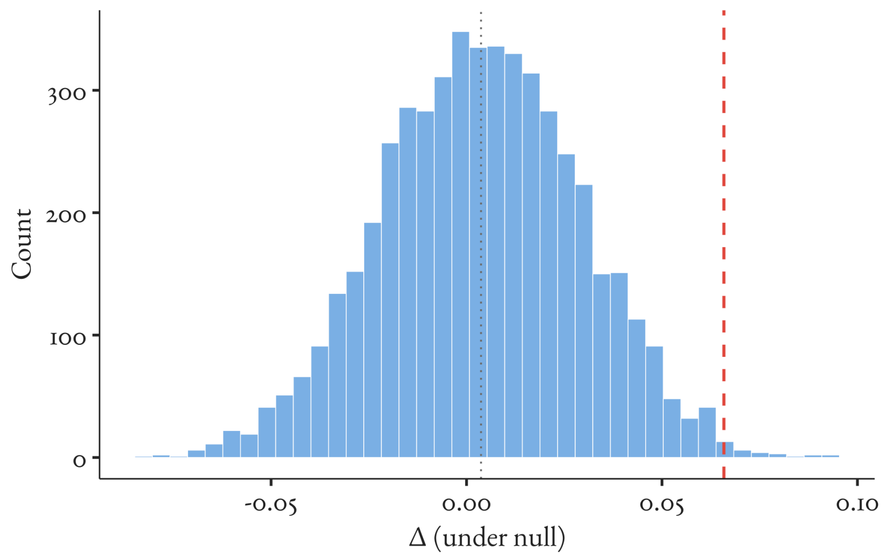


Figure 2: Constrained reference distribution for the observed Δ contrast. The histogram shows Δ values from 5,000 row- and column-preserving quasiswap scramblings of the feature matrix; the dashed vertical line marks the observed value. The figure doesn't by itself establish boundary status. It shows that the observed determinative-ward contrast is unusual under this benchmark.

The results are shown in Figure 2. For my primary comparison – using carefully matched sets of six pronouns and six compound determinatives – the observed contrast falls in the extreme positive tail of this reference distribution (tail-area summary = 0.003, Monte Carlo SE $\approx$ 0.001).

I use this tail area as a model-checking summary, not as a classical test of a sharp null hypothesis. The reference model is intentionally artificial: it asks what Δ values are typical when the instrument's row and column totals are preserved but the item–feature associations are otherwise scrambled. Under that benchmark, contrasts this determinative-ward are rare. That suggests the observed Δ isn't merely an artifact of how common or rare the coded features happen to be in English.

But two caveats apply. First, with only two reciprocals, no single contrast should carry much interpretive weight on its own. Second, my choice of which six pronouns and six determinatives to compare matters. Different selections might yield different results, a sensitivity I explore in Section 5. For that reason, I treat the quasiswap result as one reference-model check on the determinative-ward contrast rather than as a decisive test of boundary status.[5]

4 The Garden of Forking Paths

As I point out in the introduction, I've made several analytic choices: Jaccard as the primary distance, specific reference items, a particular test statistic. Each choice is a fork in the path and an opportunity for bias to accumulate. Exploring only one path risks over-interpreting a pattern that might disappear under equally reasonable alternatives.

Specification-curve analysis (Simonsohn, Simmons & Nelson 2020) addresses this directly. Instead of defending a single pipeline, I run many plausible pipelines and show all results. Here, I vary the distance metric (Jaccard, Dice, and an IDF-weighted Jaccard), the feature set (all features, or blockwise ablations such as "no morphology" and "no syntax"), and the weighting method.

The plot labels this "alpha" – a technical term for regularization. The "ridge" method keeps all features weighted, whereas "elastic05" allows the algorithm to exclude uninformative features. Each combination yields a value of Δ:

$$\Delta = \bar{d}(\text{reciprocals}, \text{pronouns}) - \bar{d}(\text{reciprocals}, \text{compound determinatives}),$$

[5] Implementation in `03_matched_set_robustness.R`; reference draws saved as CSV/RDS for reproducibility.

the mean distance from the reciprocals to pronouns minus the mean distance from the reciprocals to compound determinatives.

Figure 3: Specification curve for Δ. Each point shows the observed contrast under different combinations of weighting method (shape), distance metric (colour), and feature block exclusion (panels). Positive values indicate the reciprocals are closer to compound determinatives; negative values indicate they are closer to pronouns; zero marks the boundary.

Figure 3 shows two clear patterns. First, the sign of Δ is stable across distance metrics and weighting methods: most specifications land on the same side of zero. Second, only one block flips the sign – when morphology is *excluded* (panel "morph"), $\Delta < 0$ and the reciprocals look more pronoun-like.

With all features included (panel "none") $\Delta > 0$, and removing semantics pushes Δ further positive (more determinative-like). Removing syntax or phonology leaves small, positive contrasts near the baseline. This pattern suggests that morphological features carry most of the determinative-ward signal, whereas semantic features contribute a pronoun-ward pull; syntax and phonology are comparatively weak in this contrast.

This invariance has theoretical significance. The specification curve doesn't settle any deeper ontology of categories on its own, but it does test whether the observed interface position is robust across reasonable metrics and ablations. Δ may shift in magnitude across specifications, but its sign – indicating which side of the pronoun/compound-determinative interface reciprocals favour – remains stable across most of them. Within this measurement model, that's the signature of a stable boundary position rather than a metric artifact.

What the specification curve does *not* address is design sensitivity from the particular 6+6 matched sets used in the reference-distribution contrast; that is investigated separately by rotating the matched sets in Section 5.

5 The Importance of Comparison Groups

I had chosen specific pronouns and determinatives for comparison based on theoretical considerations, but different choices may yield different contrasts. To assess design sensitivity, I repeated the analysis 100 times, each time randomly selecting six pronouns and six compound determinatives from the eligible pools while holding all other steps fixed.

Figure 4 shows wide variation across rotations.[6] The prespecified set yields $\Delta = 0.079$; across rotations the median contrast is 0.023, with an interquartile range from 0.002 to 0.043. Three quarters of the rotations remain determinative-ward, but the effect is often much smaller than in the prespecified comparison.

With such small n, tail-area summaries are especially vulnerable to Type S (sign) and Type M (magnitude) errors (Gelman & Carlin 2014), so the more informative summary is the distribution of Δ itself rather than a tail area attached to one matched set. The substantive point is that Δ is design-sensitive to the choice of comparators: the prespecified set sits high in the rotation distribution, but many equally reasonable comparator sets yield weaker contrasts.

[6] Implementation: `03\_matched\_set\_robustness.R`; the specific canonical items are listed in `matched_subset_manifest.txt`.

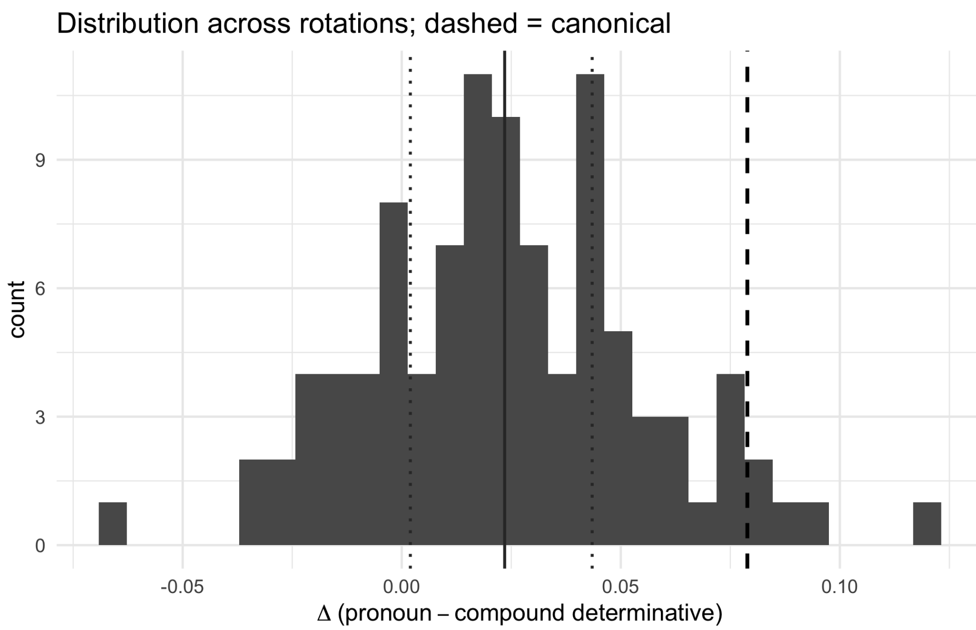


Figure 4: Sensitivity of Δ to the choice of comparison groups. The histogram shows the observed Δ values across 100 rotations of six pronoun and six compound determinative anchors. The dashed line marks the prespecified comparison set, the solid line marks the rotation median, and the dotted lines mark the interquartile range. Most rotations remain determinative-ward, but the magnitude varies substantially with the comparator set.

Accordingly, I focus on the effect itself – the distance contrast Δ – and on how Δ varies across rotations, using the rotations as a descriptive multiverse (Steegen et al. 2016) that quantifies uncertainty due to researcher degrees of freedom rather than as a filter for statistical significance.

6 Simulating the Boundary

To build intuition about what these statistical patterns mean grammatically, I constructed a simple predictive blend model. Pronouns combine feature probabilities in one way, compound determinatives in another, and a boundary item is best summarized by the blend that predicts its observed profile most accurately.

The two category profiles are estimated from the anchor items with the reciprocals held out; the exercise asks how much of each profile best predicts a reciprocal's observed properties.[7]

The simulator implements this idea with a mixture weight w that ranges from 0 to 1. When $w = 1$, the simulator generates features like a typical pronoun profile; when $w = 0$, like a typical determinative profile. For values in between, it mixes the two patterns – at $w = 0.7$, for instance, it's 70% pronoun-like and 30% determinative-like. Read w as a predictive index of boundary position, not as a literal claim about composition.

I then asked: what mixture weight would make the simulator best match the reciprocals' observed feature patterns under the same scoring rule? This is like asking: if reciprocals are intermediate between the two anchor profiles, what blend best predicts their properties?

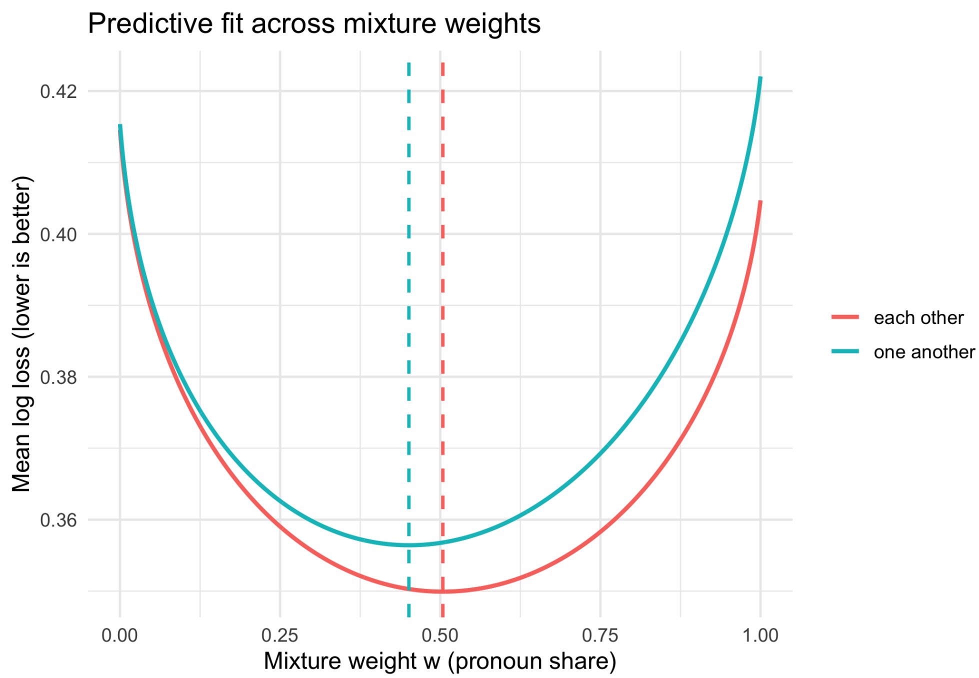


Figure 5: Predictive blending of pronoun and compound determinative profiles. Curves show predictive fit as the mixture weight w varies; dashed lines mark the best-fitting weights for each reciprocal. Both minima lie near parity, with $\hat{w} = 0.504$ for *each other* and $\hat{w} = 0.451$ for *one another*.

Figure 5 reveals the same interface from another angle. To match the properties of *each*

[7] Concretely, the category profiles are fit on conservative clear-anchor sets. For each feature, the pronoun and compound-determinative anchor sets provide smoothed Bernoulli rates. For a given weight w, the blend probability is $wp_{\text{pron}} + (1 - w)p_{\text{det}}$, and the predictive score for a reciprocal's 155-bit feature vector is its log loss under that blended profile. $\hat{w}$ is the minimizer. Implementation in `04\_weight\_calibration.R`.

other, the best blend is about 0.50 pronoun-like; for *one another*, about 0.45 pronoun-like.

This provides another lens on the same phenomenon. Whether we visualize grammatical space, measure distances, compare the observed contrast to a constrained reference distribution, or calibrate this simple blend, reciprocals consistently appear at the interface rather than clustering near either anchor profile.

Two caveats apply. First, reciprocals aren't literally probability mixtures of two kinds; the simulator provides an interpretable, one-number summary of where they sit relative to the anchor profiles. Second, the blend is estimated on the same measurement instrument as the other analyses, so it's another internal check rather than an independent source of evidence.

7 Checking the Instruments

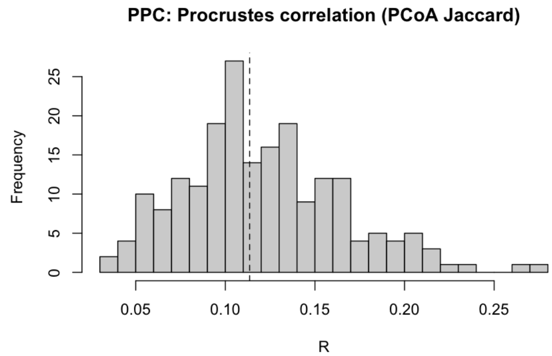
Throughout these analyses, I have treated the feature matrix as a measurement model – but what if the measurements themselves are problematic? The first question isn't whether the richest model fits. It's whether the workflow earns trust step by step by distinguishing clear anchors and preserving basic structure before it's asked to interpret an ambiguous case.

The checks reported here start one step later. I fit a simple model that predicts each binary feature from the item's category label with modest adjustment terms and weak regularization, then draw replicated datasets from the fitted model.[8]

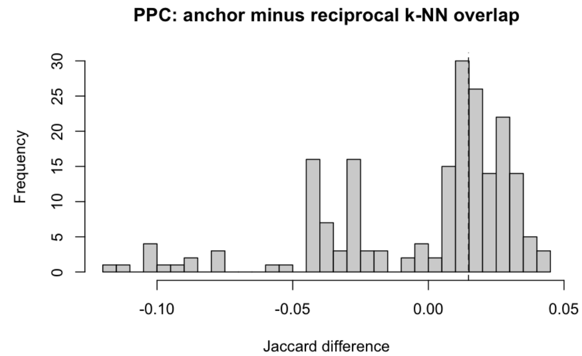
That staged buildup matters because a complicated fit can look plausible while still hiding an implementation error.

I ask two questions. Global geometry: do replicated datasets preserve the large-scale arrangement (which items cluster with which) under the same ordination? Local neighbourhoods: do they preserve nearest-neighbour relations around anchors?

[8] Concretely, for feature j on item i, $y_{ij} \sim \mathrm{Bernoulli}(\pi_{ij})$ with $\operatorname{logit} \pi_{ij} = \alpha_j + \beta_j \mathbf{1}\{c_i = \text{pronoun}\}$ and ridge regularization on β_j; draws are parametric bootstrap replicates (posterior draws if a Bayesian fit's used). Reciprocals are held out wherever predictions for them are reported.



(a) Predictive check for global structure. Histogram: distribution of the Procrustes statistic under replicated datasets; vertical line: observed statistic.



(b) Predictive check for local structure. Histogram: distribution of k-NN overlap around anchors (k fixed across runs); vertical line: observed overlap.

Figure 6: Predictive checks. (a) Global structure: distribution of Procrustes fit statistics across replicated datasets (histogram) with the observed statistic marked (vertical line). (b) Local structure: distribution of nearest-neighbour overlaps under replication, with the observed overlap marked.

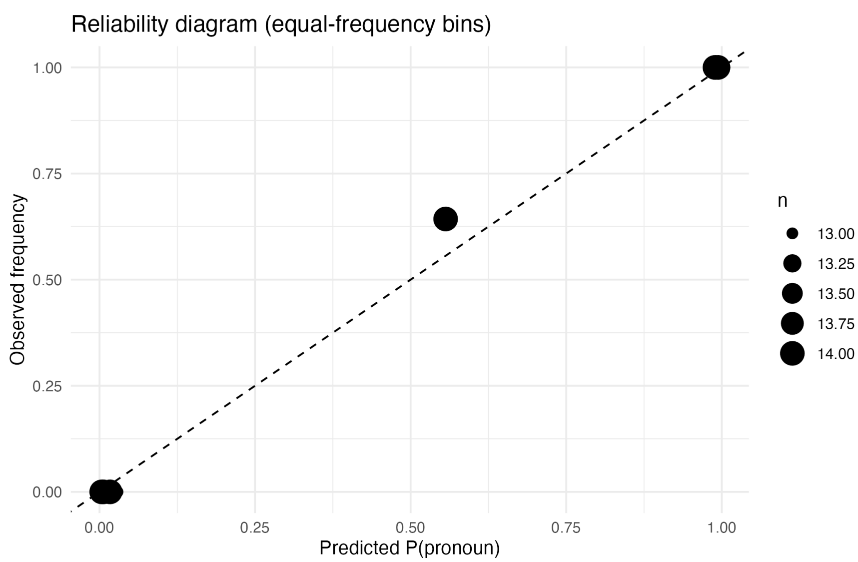


Figure 7: Reliability diagram under cross-validation, following Niculescu-Mizil & Caruana (2005). Predicted probabilities (x-axis) track observed frequencies (y-axis), showing approximate calibration: when the model predicts 70% probability for the pronoun label, about 70% of held-out items are pronouns.

The model passes these basic predictive checks (Figures 6a–7). Replicates maintain reasonable global geometry and preserve about 65.5% of local neighbourhood structure around anchors.

The reliability diagram shows that when the classifier assigns, say, 70% probability to the pronoun label, the realized frequency on held-out items is near that level – probabilities are roughly calibrated rather than systematically over- or under-confident.[9]

Perfect reproduction is neither expected nor desired. The checks indicate that the measurement model carries real signal about categorical structure: clear pronouns and clear compound determinatives are distinguishable under the same machinery, which makes the reciprocals' ambiguity informative rather than an artefact of a broken instrument.

8 What We Learn About Grammatical Boundaries

Taken together, the results instantiate the HPC-derived expectations. (E1) holds: the sign of the distance contrast is stable across metrics and penalties and is only modest in magnitude; reference-distribution extremity is design-sensitive but the qualitative boundary diagnosis persists across rotations. (E2) holds: morphology contributes most of the determinative-ward pull while semantics contributes a pronoun-ward pull, with syntax and phonology comparatively weak in this contrast. (E3) holds: the same machinery separates anchors and yields calibrated supervised probabilities for them.

(E4) holds: the predictive blend places both reciprocals near parity. (E5) holds for the prespecified anchor contrast: the quasiswap reference distribution that preserves row and column totals rarely reproduces a contrast this determinative-ward, while the rotation analysis shows that its tail-area extremity is design-sensitive. Under an HPC reading, this pattern – stable ambiguity with cross-dimensional tension and clean anchor behaviour – is what a stable boundary position looks like within this measurement model.

This pattern is compatible with the homeostatic property-cluster view of categories. The reciprocals exhibit stable ambiguity with cross-dimensional tension rather than a decisive recategorization or a failure of measurement. Morphology supplies most of the determinative-

[9]For the reciprocals, the classifier assigns near-chance pronoun probabilities (0.485 for *each other*, 0.467 for *one another*); masked predictive log-likelihood is identical under either labelling (-54.240), and per-item cross-entropy is 0.371 versus a chance baseline $\ln 2 \approx 0.693$.

ward pull; semantics supplies a smaller pronoun-ward pull; syntax and phonology contribute little to this particular contrast. Within this instrument, additional statistical machinery is unlikely to yield a clean assignment because the conflict is real rather than an artefact of a specific pipeline; changing the conclusion would require different measurements or external evidence, not a different threshold.

The methodological lesson is general. Boundary items should be expected – and tested – to show stable ambiguity under reasonable analytic choices. Single diagnostics or cherry-picked features force binary decisions; comprehensive profiles, prespecified lenses, and summaries that propagate design uncertainty instead map the topology of grammatical space, including its interfaces, and make clear what is robust within the chosen measurement model.

9 Discussion

On an HPC view, success at a boundary isn't a decisive recategorization but a stable pattern of ambiguity accompanied by cross-dimensional conflict. That is the profile reciprocals exhibit here.

Across lenses that make different invariance assumptions yet interrogate the same measurements, reciprocals sit at the edge of the pronoun region in ordinations, show mixed nearest neighbours, and attract near-chance supervised probabilities with tied predictive fit (0.485, 0.467; log-likelihood -54.240 under either labelling).

They also display a blockwise split, with morphology leaning determinative-ward,[10] semantics leaning pronoun-ward, and syntax and phonology contributing little.

Read through the HPC lens, the mixture calibration reproduces decisive separation for anchors while placing reciprocals at the boundary with near-parity placement, consistent with the ordination, distance, reference-distribution, and supervised results. Because all lenses interrogate the same measurements, agreement is internal robustness rather than independent convergence.

[10] This likely reflects pronouns' tendency to inflect for case, which reciprocals resist.

The blockwise pattern has a natural mechanistic interpretation. On an HPC view, different homeostatic mechanisms sustain different portions of the property cluster. Morphological realization rules and agreement systems – mechanisms that maintain the determinative basin – account for the morphology-ward pull. Interpretive mechanisms governing reference tracking, binding, and argument structure – mechanisms that maintain the pronoun basin – account for the semantics-ward pull. The cross-dimensional tension is not a quirk of the feature matrix; it is consistent with distinct mechanisms exerting opposed pressures on the reciprocals' categorical location.

The findings are consistent with multiple theoretical frameworks that predict stable ambiguity at category boundaries. What distinguishes the HPC interpretation is the cross-dimensional tension: different feature families pulling in different directions.

The paper's direct claim is measurement-level rather than metaphysical. In this CGEL-informed feature space, reciprocals occupy a stable position at the pronoun/compound-determinative interface.

What our instruments reveal is proximity to that interface: near-chance classifier probabilities, near-parity mixture weights, and cross-dimensional tension. The situation is analogous to a microscope at its resolution limit: the instrument localizes an interface zone even when it doesn't settle every deeper question about underlying ontology. The 155-feature matrix has an analogous resolution floor – and reciprocals fall within it.

An HPC interpretation makes sense of this pattern because it predicts partially distinct mechanisms maintaining partially overlapping bundles of properties. On that reading, stable interface placement combined with systematic cross-dimensional conflict is what we should expect when the bundles diverge at the edge.

This interpretation also clarifies what the results don't show. The analyses don't accumulate independent evidential weight; all procedures reuse the same hand-coded matrix. Agreement across lenses therefore counts as internal robustness against method-specific artifacts rather than as independent convergence. Nor do the numbers license recategorization.

On the predeclared matched set the observed contrast sits far into the positive tail of the quasiswap reference distribution (tail-area summary = 0.003), but across rotated comparison sets the effect size varies substantially. Under an HPC reading, those very features – modest directionality, design sensitivity, and stability of the qualitative diagnosis across reasonable specifications – are what an interface position is expected to look like given the target population of two types.

The fusion-of-functions architecture provides a complementary grammatical backdrop (Payne, Huddleston & Pullum 2007). If category and function are distinct primitives and fused determiner–head structures are grammatically real, then morphologically complex items can be determinatives by category while serving fused functions in NP.

That architecture predicts pressure toward categorical edges when different mechanisms sustain partially conflicting property bundles. The present measurements are consistent with such pressure for the reciprocals: they behave pronoun-like on some feature classes and determinative-like on others. At the same time, retaining the canonical CGEL pronoun analysis as the prior default is methodologically appropriate; the data here don't overturn it, and the framework shown here makes explicit what kinds of evidence would do so.

Seen more broadly, the study's contribution is methodological and interpretive rather than classificatory. It operationalizes a way to examine boundary phenomena that avoids selective diagnostic shopping (Croft 2001): prespecify a small set of sensible lenses; interpret agreement as internal robustness on one dataset; make prior theoretical commitments explicit; and report the full specification surface rather than only favourable summaries. Inconclusive tests cease to be mere failures when they co-occur with cross-dimensional conflict and a stable ambiguity pattern; they become data about the structure of categories if those categories are homeostatic clusters (Miller 2021).

9.1 Limitations, scope conditions, and disconfirmation

All findings are conditional on a single, theory-informed, hand-coded binary measurement model. Features are binarized, single-coder, and CGEL-dependent; this raises construct-validity and reliability concerns. I partly offset these with blockwise analyses and metric choice motivated by ordination, but stronger guarantees would require multi-annotator coding with reliability estimates or corpus-derived distributional features that reduce theory-dependence.

Target scarcity constrains inferential resolution: English has two reciprocal types, so any single reference-distribution contrast between reciprocals and comparators is inherently fragile, and supervised categorization produces only two probabilities of primary interest. Multiplicity and researcher degrees of freedom are addressed by prespecifying a small, theory-led set of lenses and reporting all analyses. Specification choices nonetheless remain scope conditions on the results.

External validity is restricted to English and to the CGEL-based inventory; cross-linguistic generalization is a matter for new data.

The framework also supplies clear disconfirmation criteria for this case. Outcomes that would count against the boundary diagnosis include: alignment of morphology, syntax, semantics, and phonology on the same pole; calibrated supervised probabilities near 1.0 under one labelling together with superior predictive fit; disappearance of ambiguity under reasonable distance metrics or regularization choices; and stability of those outcomes under feature-subset resampling. None of these criteria is met here.

9.2 Implications and outlook

The present case is intended to illustrate how to study boundary phenomena responsibly when categories are treated as homeostatic clusters. The objective isn't to force a verdict on reciprocals but to refine the workflow that registers boundary behaviour on the measurement model already in use.

First, the workflow should characterize dependence rather than "repair" individual codings in a large, noisy matrix. What matters is which parts of the measurement model underwrite the boundary diagnosis.

Influence diagnostics for distance-based summaries (feature leverage on Jaccard neighbourhoods), blockwise ablations, and adversarial recoding of plausible high-leverage features jointly map an internal robustness surface: which modest, theory-motivated perturbations do and don't alter the qualitative outcome. This keeps the focus on stability of ambiguity rather than on accumulating additional evidence.

Second, the workflow should calibrate against control items. Clear pronouns and clear compound determinatives should behave as anchors under the same lenses: high-confidence, aligned outcomes for anchors and ambiguity for reciprocals. Reporting anchor performance turns robustness from a general claim into a concrete diagnostic. If ambiguity appears for anchors, the pipeline is uninformative; if it localizes to reciprocals, the boundary reading gains credibility without appealing to data beyond the current inventory.

Third, the submission package should make the measurement instrument easier to inspect. An anonymized supplement accompanying this submission provides the full 155-property matrix, the feature-block map, the item-level inventory annotations, the anchor manifests, the key analysis scripts, the matched-set manifest, the quasiswap reference draws for the canonical contrast, and derived robustness summaries. That lets readers see exactly which coding decisions do and don't move the interface position.

Fourth, predictive comparison should remain in its place. If PSIS-LOO enters the workflow, it should come after posterior predictive checks and with Pareto-k diagnostics reported. Here its role is secondary because the main question is boundary location, not model ranking.

Fifth, token-level evidence, where available, should be used as a diagnostic of boundary dispersion rather than as a route to a type-level decision. Treat the lexeme as a distribution over constructions and modification profiles, and compare simple, predeclared dispersion summaries (e.g. entropy over syntactic frames; divergence from anchor frame distributions) to the anchors.

Under a cluster view, a boundary item is expected to show broader, more heterogeneous contextual support than clear-category anchors. A hierarchical analysis can pool information without conflating multiple types; the goal is to quantify dispersion as a property of a boundary item, not to recast token counts as votes.

These steps remain within English and within the existing theoretical commitments. They raise evidential quality by making the dependence structure of the present results explicit, by checking behaviour against anchors, by making the instrument itself easier to inspect, and – where feasible – by quantifying boundary-like dispersion in usage. That set of implications is congruent with the study's aim: to show how boundary cases can be investigated rigorously under an HPC lens without presuming that a decisive recategorization is either necessary or available.

Finally, one extension testable with independent data is to collect acceptability judgments on constructions that differentiate pronouns from determinatives, predicting that reciprocals show higher judgment variance than clear anchors in exactly those contexts. If distance-to-boundary (as measured here) predicts judgment variance, that would confirm the stable-ambiguity pattern reflects proximity to a discrete threshold rather than inherent category fuzziness.

While this analysis focuses on English reciprocals within established theoretical frameworks, field linguists face analogous challenges with boundary phenomena in understudied languages. The core insight – that stable ambiguity can be more informative than forced categorization – applies even without elaborate statistical machinery.

A field linguist documenting, say, Oceanic classifiers that show both nominal and verbal properties, might adapt this approach by: (1) documenting the full range of conflicting properties rather than privileging select diagnostics, (2) explicitly comparing to clear anchors in each category, and (3) acknowledging boundary status in reference materials rather than forcing binary decisions. The statistical framework presented here may be impractical for initial documentation, but the principle of transparent, multi-lens investigation of boundary phenomena remains valuable.

10 Conclusion

If grammatical categories are homeostatic property clusters (Miller 2021), we need methods that can register stable interface positions – especially when we're working with tiny samples. This paper offers one such method, demonstrated on English reciprocals but designed for broader application.

The core insight is simple: stop trying to force binary decisions on boundary phenomena. Instead, measure whether the diagnostic ambiguity itself is stable. Across every lens I applied – distance measurements, statistical classifiers, quasiswap reference-model checks, specification curves, and predictive blend models – reciprocals consistently occupied the pronoun/compound-determinative interface. The best-fitting mixture weights keep them near parity between the anchor profiles, and this location barely budges under reasonable changes in metric and feature family.

This stability of diagnostic ambiguity is informative. It tells us that reciprocals aren't just hard to classify due to noisy data or poor methods – they sit stably at the interface where this instrument loses resolution. Their morphology pulls them toward the compound determinatives of Payne, Huddleston & Pullum (2007), their semantics toward pronouns. Within this measurement model, additional machinery is unlikely to overturn the interface reading without new measurements or external evidence. But this remains internal robustness on a single instrument, not independent convergence across datasets.

For linguists facing similar problems, the workflow can be summarized as follows:

1. Build comprehensive feature profiles (don't cherry-pick diagnostics)

2. Show representative diagnostics before summarizing them quantitatively

3. Let visualization guide distance metrics (use ordination first)
4. Compare key contrasts to constrained reference distributions (especially with small n)
5. Vary specifications systematically (show all results)
6. Calibrate against clear cases (verify known structure)
7. Report effect distributions, not only tail areas, when comparator choice matters

The same two limitations I noted initially remain. Everything depends on the hand-coded feature matrix – transparent and theory-grounded, but still one person's coding. And with only two reciprocals, we're measuring position in calibrated space rather than making population inferences – the analyses characterize where these specific items fall, not whether reciprocals as a class differ from other categories. These constraints are inherent to the problem, not failures of method.

But within these constraints, the approach delivers something valuable: a way to study boundary phenomena that's both rigorous and honest about uncertainty. We don't need to pretend every word fits neatly into a predetermined box. Some words, like reciprocals, occupy stable interface positions, and that is worth documenting carefully.

The methods transfer directly to other small-n categorization problems. Whether you're investigating modal auxiliaries, discourse particles, or any other small group with uncertain categorical status, this same approach applies. The procedures are documented, and the logic is transparent.

The reciprocals remain categorized as pronouns in standard grammars including CGEL, and nothing here overturns that categorization. What I have shown is narrower: within this CGEL-informed measurement model, they occupy a stable peripheral position at the pronoun/compound-determinative interface. That's not a failure of categorization; it's a more explicit characterization of where the difficult cases sit.

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